

PROJECT 01

Self-Tighten Gripper for Lalan Tensile Testing Machine

Effective Engineering & Automation Pvt Ltd (Client: LALAN Rubber Co.)

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1. ABSTRACT

This project involved the complete mechanical design and fabrication of a self-tightening gripper mechanism to replace the existing screw-actuated clamping fixture on the LALAN Rubber tensile testing machine. The gripper exploits an eccentric pin kinematics principle to generate an automatic clamping force proportional to the applied tensile load, eliminating the need for manual screw adjustment between test cycles. The final assembly achieved a 20 mm reduction in gripper footprint, significantly reduced sample changeover time, and demonstrated repeatable clamping performance across the required test-force range.

2. TECHNICAL BACKGROUND & PROBLEM STATEMENT

Tensile testing of elastomeric and fabric specimens requires a gripper that applies sufficient clamping force to prevent specimen slip while avoiding damage to the material under test. The original screw-driven clamp at LALAN required manual tightening before each test cycle, creating operator fatigue, cycle-time variability, and risk of inconsistent grip force. The requirement was to engineer a replacement that (a) self-tightens under applied tensile load, (b) occupies a smaller axial footprint within the machine crosshead, and (c) allows rapid specimen insertion without tools.

3. TECHNICAL OBJECTIVES

1. Design a self-tightening clamping mechanism utilising eccentric pin kinematics to generate grip force proportional to tensile load.
2. Reduce the overall gripper axial footprint by a minimum of 15 mm compared to the existing screw-driven assembly.
3. Achieve specimen changeover without the use of hand tools, reducing cycle time.
4. Engineer the design for manufacture using conventional workshop processes (turning, milling) with a cost-efficient material selection.
5. Apply surface treatments (nickel plating, knurling) to achieve corrosion resistance and adequate coefficient of friction on gripping surfaces.

4. SYSTEM ARCHITECTURE & DESIGN OVERVIEW

The self-tightening mechanism operates on an eccentric pin kinematic principle. The gripping jaw is mounted on an eccentrically positioned pivot pin; as axial tensile load is transmitted through the specimen, the resulting moment about the eccentric pin axis drives the jaw to rotate into a tighter clamping position. The clamping force is therefore self-regulating — it increases proportionally with applied load, preventing slip without over-clamping at low loads. A return spring resets the jaw to the open (specimen-insertion) position when tensile load is removed. The assembly interfaces with the existing crosshead T-slot geometry, requiring no machine modification.

5. TECHNICAL SPECIFICATIONS

Parameter	Value / Description	Unit / Notes
Operating principle	Eccentric pin self-tightening kinematics	—
Primary material	Mild steel (MS rod, MS plate)	Structural members
Surface treatment — grip	Knurling, straight pattern	Increases friction coefficient
Surface treatment — body	Electroless nickel plating	Corrosion resistance
Footprint reduction	20	mm (vs. original gripper)
Actuation	Passive (load-driven) — no external actuator	—
Specimen type	Elastomeric / fabric strip	Per LALAN test requirements
CAD platform	SolidWorks 2022	3D model + 2D fabrication drawings

6. KEY COMPONENTS & BILL OF MATERIALS

#	Component	Specification	Function
1	MS Rod	Ø[TBF] mm × [TBF] mm, grade S235	Gripper jaw and body
2	MS Plate	t = [TBF] mm, grade S235	Mounting bracket and base plate
3	Eccentric Pin	Ø[TBF] mm, hardened steel	Self-tightening pivot element
4	Return Spring	Compression spring, k = [TBF] N/mm	Jaw reset to open position
5	Pin Clip (circlip)	DIN 471, Ø[TBF] mm	Axial retention of pivot pin
6	Nuts & Bolts	M[TBF] ISO metric, hex-head	Assembly fasteners
7	Knurled Insert	MS, straight knurl, 0.8 mm pitch	Friction grip face

7. ENGINEERING METHODOLOGY & DESIGN PROCESS

The design process followed a structured mechanical engineering methodology. Initial task analysis established the kinematic requirements for self-tightening under axial load. A literature review of existing self-tightening fixture principles (eccentric, wedge, toggle) was conducted to identify the most suitable mechanism. Conceptual sketches were developed and reviewed with the supervising design engineer (Eng. Suneth). A parametric SolidWorks 3D model was constructed using actual crosshead measurements to verify spatial fit. The design underwent three iterative refinement cycles incorporating feedback from the design engineer before fabrication drawings were released to the workshop. Post-fabrication, a functional test protocol was executed including load cycling to verify self-tightening behaviour and slip resistance.

8. ENGINEER'S CONTRIBUTION

- Conducted kinematic analysis and mechanism research to select the eccentric pin self-tightening principle.
 1. Produced hand-sketched conceptual designs and presented for design review and approval.
 1. Constructed parametric 3D SolidWorks model from physical machine measurements.
 1. Iterated the design through three revision cycles incorporating design engineer corrections.
 1. Prepared complete set of 2D fabrication drawings (individual parts and assembly) for workshop submission.
 1. Supervised and participated in component fabrication (turning, milling, drilling) and assembly.
 1. Executed functional testing protocol including tensile load cycling and grip-slip assessment.
 1. Identified and resolved post-plating grip force degradation through surface re-evaluation.
 1. Prepared full technical documentation including design rationale and test records.

9. TESTING & VALIDATION

Functional validation was conducted in three phases: (1) Dry assembly test — mechanism kinematics verified manually by applying simulated load via workshop vice, confirming eccentric pin engagement and jaw closure. (2) On-machine load test — specimen inserted and tensile load applied incrementally; grip slip force measured at each load step to confirm self-tightening characteristic. (3) Repeatability test — 20 consecutive specimen insertion and removal cycles performed by operator, measuring changeover time and confirming consistent jaw reset. Nickel plating was applied post-assembly and grip tests repeated to quantify friction reduction; a knurl geometry revision was subsequently incorporated.

10. TECHNICAL CHALLENGES & ENGINEERING SOLUTIONS

Post-nickel-plating, the electroless nickel coating (typically 25–50 μm deposit) filled the knurl microgeometry on the grip faces, reducing the effective friction coefficient and degrading specimen retention at moderate tensile loads. This was diagnosed by comparative grip-slip load measurement before and after plating. The solution involved re-machining the knurl after plating to restore surface texture depth, followed by re-validation of grip performance at the specified test load range.

11. RESULTS & PERFORMANCE METRICS

The redesigned gripper assembly delivered: (1) 20 mm reduction in axial footprint within the crosshead, allowing greater specimen gauge length utilisation. (2) Elimination of manual screw tightening — specimen changeover time reduced to a single-hand insertion action. (3) Consistent self-tightening performance confirmed across 20 repeat test cycles with no observed specimen slip at design load. (4) Cost-efficient fabrication from standard mild-steel stock and in-house workshop processes.

12. CONCLUSIONS & FUTURE RECOMMENDATIONS

The eccentric pin kinematic principle proved an effective and low-cost mechanism for passive self-tightening under axial tensile load. Future recommendations include: (a) conducting a formal FEA

stress analysis on the eccentric pin under peak load to establish a safety factor margin; (b) investigating a hardened steel or ceramic-coated insert for the grip face to provide durable friction without requiring post-plating rework; (c) evaluating a quick-release pin configuration to further reduce changeover time.

13. TOOLS, SOFTWARE & EQUIPMENT

Tool / Software	Category	Application in Project
SolidWorks 2022	CAD Software	3D parametric modelling, assembly design, and 2D fabrication drawing generation
Lathe Machine	Machine Tool	Turning of eccentric pin, jaw body, and rod components to tolerance
Milling Machine	Machine Tool	Milling of MS plate brackets, slot cutting, and face milling
Knurling Tool	Hand / Machine Tool	Diamond and straight knurl application to grip faces
Electroless Nickel Plating	Surface Treatment	Corrosion-resistant coating on structural steel components
Vernier Caliper (0.02 mm)	Metrology	Dimensional inspection of machined components
Workshop Hand Tools	Tooling	Assembly, fitting, and adjustments

14. IMAGES, TECHNICAL DRAWINGS & CAD SCREENSHOTS

[Insert SolidWorks renders, fabrication drawings, assembly photos, and annotated images here.]

15. REFERENCES, DEMO LINKS & APPENDICES

[Insert Google Drive / YouTube demo link, reference standards, datasheets, or appendices here.]